

Time-stamp: < 09 oct 2024 11:55 *Philippe Queinnec* >

Algorithme d'exclusion mutuelle à base de *jeton* – version avec un tableau pour *jeton*

EXTENDS *Naturals*, *FiniteSets*

CONSTANT *N*

ASSUME $N \in \text{Nat} \wedge N > 1$

$\text{Processus} \triangleq 0 \dots N - 1$

$\text{Hungry} \triangleq \text{"H"}$

$\text{Thinking} \triangleq \text{"T"}$

$\text{Eating} \triangleq \text{"E"}$

VARIABLES

etat,

jeton

$\text{TypeOK} \triangleq$

$\square(\wedge \text{etat} \in [\text{Processus} \rightarrow \{\text{Hungry}, \text{Thinking}, \text{Eating}\}]$
 $\wedge \text{jeton} \in [\text{Processus} \rightarrow \text{BOOLEAN}])$ *jeton est un tableau de booléens*

un processus max a le jeton

$\text{ExclMutuelle} \triangleq$

$\square(\forall i, j \in \text{Processus} : (\text{etat}[i] = \text{Eating} \wedge \text{etat}[j] = \text{Eating}) \Rightarrow (i = j))$

un processus qui demande obtiendra plus tard

$\text{VivaciteIndividuelle} \triangleq$

$\forall i \in \text{Processus} : (\text{etat}[i] = \text{Hungry} \leadsto \text{etat}[i] = \text{Eating})$

si un processus demande, un processus obtiendra plus tard (pas forcément le même)

$\text{VivaciteGlobale} \triangleq$

$(\exists i \in \text{Processus} : \text{etat}[i] = \text{Hungry}) \leadsto (\exists j \in \text{Processus} : \text{etat}[j] = \text{Eating})$

$\text{JetonVaPartout} \triangleq$

$\forall i \in \text{Processus} : \square \diamond (\text{jeton}[i])$

Le jeton est un tableau, on doit vérif qu'il est unique

$\text{JetonUnique} \triangleq$

$\square(\forall i, j \in \text{Processus} : (\text{jeton}[i] \wedge \text{jeton}[j]) \Rightarrow (i = j))$

$\text{Sanity} \triangleq$

$\square(\forall i \in \text{Processus} : \text{etat}[i] = \text{Eating} \Rightarrow \text{jeton}[i])$

$\text{Init} \triangleq$

$$\begin{aligned} & \wedge \text{etat} = [i \in \text{Processus} \mapsto \text{Thinking}] \\ & \wedge \exists j \in \text{Processus} : \text{jeton} = [i \in \text{Processus} \mapsto i = j] \text{ génie} \end{aligned}$$

$$\begin{aligned} \text{demander}(i) & \triangleq \\ & \wedge \text{etat}[i] = \text{Thinking} \\ & \wedge \text{etat}' = [\text{etat} \text{ EXCEPT } ![i] = \text{Hungry}] \\ & \wedge \text{UNCHANGED } \text{jeton} \end{aligned}$$

$$\begin{aligned} \text{entrer}(i) & \triangleq \\ & \wedge \text{etat}[i] = \text{Hungry} \\ & \wedge \text{jeton}[i] \\ & \wedge \text{etat}' = [\text{etat} \text{ EXCEPT } ![i] = \text{Eating}] \\ & \wedge \text{UNCHANGED } \text{jeton} \end{aligned}$$

$$\begin{aligned} \text{sortir}(i) & \triangleq \\ & \wedge \text{etat}[i] = \text{Eating} \\ & \wedge \text{etat}' = [\text{etat} \text{ EXCEPT } ![i] = \text{Thinking}] \\ & \quad \text{“déplacer” le jeton de } i \text{ à } (i+1)\%N \\ & \wedge \text{jeton}' = [\text{jeton} \text{ EXCEPT } ![i] = \text{FALSE}, ![(i+1)\%N] = \text{TRUE}] \end{aligned}$$

$$\begin{aligned} \text{bouger}(i) & \triangleq \\ & \wedge \text{jeton}[i] \\ & \wedge \text{etat}[i] \neq \text{Eating} \\ & \quad \text{“déplacer” le jeton de } i \text{ à } (i+1)\%N \\ & \wedge \text{jeton}' = [\text{jeton} \text{ EXCEPT } ![i] = \text{FALSE}, ![(i+1)\%N] = \text{TRUE}] \\ & \wedge \text{etat}[i] \neq \text{Hungry} \\ & \wedge \text{UNCHANGED } \text{etat} \end{aligned}$$

$$\begin{aligned} \text{Next} & \triangleq \\ & \exists i \in \text{Processus} : \\ & \quad \vee \text{demander}(i) \\ & \quad \vee \text{entrer}(i) \\ & \quad \vee \text{sortir}(i) \\ & \quad \vee \text{bouger}(i) \end{aligned}$$

$$\begin{aligned} \text{Fairness} & \triangleq \forall i \in \text{Processus} : \\ & \quad \wedge \text{WF}_{\langle \text{etat}, \text{jeton} \rangle}(\text{entrer}(i)) \\ & \quad \wedge \text{WF}_{\langle \text{etat}, \text{jeton} \rangle}(\text{sortir}(i)) \\ & \quad \wedge \text{WF}_{\langle \text{etat}, \text{jeton} \rangle}(\text{bouger}(i)) \end{aligned}$$

$$\begin{aligned} \text{Spec} & \triangleq \\ & \wedge \text{Init} \\ & \wedge \Box[\text{Next}]_{\langle \text{etat}, \text{jeton} \rangle} \\ & \wedge \text{Fairness} \end{aligned}$$